

For trigonal symmetry,

$$H_{cf} = \underline{B_2^0 O_2^0} + \underline{B_4^0 O_4^0} + \underline{B_4^2 O_4^2} + B_2^0 \cancel{O_2^2} + B_4^2 \cancel{O_4^2} + B_4^4 \cancel{O_4^4}$$

	$1\frac{1}{2}$	$1\frac{1}{2}$	$1\frac{3}{2}$	$1\frac{3}{2}$	$1\frac{3}{2}$	$1\frac{3}{2}$
$\langle \frac{5}{2} $	$10B_2^0 + 60B_4^0$				$3\sqrt{10} B_4^2$	
$\langle \frac{1}{2} $		$-8B_2^0 + 120B_4^0$				
$\langle -\frac{3}{2} $			$-2B_2^0 - 180B_4^0$			
$\langle -\frac{5}{2} $				$10B_2^0 + 60B_4^0$		
$\langle \frac{1}{2} $	$3\sqrt{10} B_4^2$				$-8B_2^0 + 120B_4^0$	
$\langle \frac{3}{2} $						$-2B_2^0 - 180B_4^0$

* Key step is to reorder the kets st. we can extract a "B" matrix that looks like $\begin{vmatrix} B & 0 \\ 0 & 0 \end{vmatrix}$

For D_4 symmetry: $H_{cf} = B_2^0 O_2^0 + B_4^0 O_4^0 + B_4^2 O_4^2 + B_2^0 \cancel{O_2^2} + B_4^2 \cancel{O_4^2} + B_4^4 \cancel{O_4^4}$

	$1\frac{1}{2}$	$1\frac{1}{2}$	$1\frac{3}{2}$	$1\frac{3}{2}$	$1\frac{3}{2}$	$1\frac{3}{2}$
$\langle \frac{5}{2} $	$10B_2^0 + 60B_4^0$		$12\sqrt{5} B_4^2$			
$\langle \frac{1}{2} $		$-8B_2^0 + 120B_4^0$				
$\langle -\frac{3}{2} $	$12\sqrt{5} B_4^2$		$-2B_2^0 - 180B_4^0$			
$\langle -\frac{5}{2} $				$10B_2^0 + 60B_4^0$		$12\sqrt{5} B_4^2$
$\langle \frac{1}{2} $					$-8B_2^0 + 120B_4^0$	
$\langle \frac{3}{2} $				$12\sqrt{5} B_4^2$		$-2B_2^0 - 180B_4^0$

↖ here already in form $\begin{vmatrix} B & 0 \\ 0 & 0 \end{vmatrix}$

where B

$$= \begin{bmatrix} 10B_2^0 + 60B_4^0 & 0 & 12\sqrt{5}B_4^2 \\ 0 & -8B_2^0 + 120B_4^0 & 0 \\ 12\sqrt{5}B_4^2 & 0 & -2B_2^0 - 180B_4^0 \end{bmatrix}$$